COFINITE ZEROS OF HIGH DERIVATIVES

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Abstract. We construct a nonzero transcendental entire function such that every nonempty open subset of the complex plane contains a zero of every sufficiently high derivative; equivalently, the union of the zero sets along every infinite increasing sequence of derivative orders is dense. The construction is probabilistic and uses a bounded-coefficient Fock series. A saddle estimate, a one-coordinate small-ball bound, and Jensen's formula give summable outer probabilities for zero-free disks. The resulting function satisfies the explicit growth bound $|f(z)| \le \sqrt{2}\exp(|z|^2)$ and therefore also supplies a counterexample to a 1973 theorem of Boas and Reddy as printed. A machine-checked Lean 4 formalization verifies the existence theorem, the growth bound, and their supporting lemmas.

1. Statement and reduction

For an entire function f and a nonempty open set $U \subset \mathbb{C}$, put $A_U(f) = \{n \in \mathbb{N} : f^{(n)} \text{ has a zero in } U\}$. We begin with the elementary quantifier reduction that will be used at the end.

Lemma 1.1. *For an entire function f, the following assertions are equivalent.*

(1) *For every strictly increasing sequence $(n_k)_{k\ge 0}$ of nonnegative integers, the set*
$$\bigcup_{k\ge 0}\{z \in \mathbb{C} : f^{(n_k)}(z) = 0\}$$
is dense in $\mathbb{C}$.

(2) *For every nonempty open $U \subset \mathbb{C}$, the set $A_U(f)$ is cofinite in $\mathbb{N}$, meaning that its complement is finite.*

Proof. Assume (2), fix U, and let (n_k) be strictly increasing. Since (n_k) is unbounded, some n_k lies beyond the finite exceptional set $\mathbb{N} \setminus A_U(f)$; hence the displayed union meets U. Conversely, if $\mathbb{N} \setminus A_U(f)$ is infinite, enumerate it increasingly as (n_k). Every zero set of $f^{(n_k)}$ then misses U, so their union is not dense. □

Theorem 1.2. *There is a nonzero nonpolynomial entire function f such that, for every nonempty open set $U \subset \mathbb{C}$, there exists $N(U)$ with*
$$n \ge N(U) \quad\Longrightarrow\quad f^{(n)} \text{ has a zero in } U.$$
The same function can be chosen to satisfy
$$|f(z)| \le \sqrt{2}\exp(|z|^2) \qquad (z \in \mathbb{C}). \tag{1}$$
Consequently, the zero sets of the derivatives indexed by every strictly increasing sequence have dense union.

Here and below, "transcendental entire" means entire and nonpolynomial. This answers the intended transcendental form of Erdős Problem #906 [1]. The final assertion follows from Lemma 1.1; the rest of the paper proves the cofinite assertion and the growth bound.

2. The random Fock series

Let $\xi_0, \xi_1, \ldots$ be independent random variables, each uniformly distributed on the closed unit disk $\overline{\mathbb{D}}$. Thus normalized planar area is their common law. This is a bounded random analytic series in the sense of the classical theory; see, for example, [6]. Define
$$f(z) = \sum_{k=0}^{\infty} \xi_k \frac{z^k}{\sqrt{k!}}. \tag{2}$$

Since $|\xi_k| \le 1$, the series converges normally on every compact subset of $\mathbb{C}$, for every sample. It therefore defines an entire function and may be differentiated termwise. For $n \ge 0$ write $F_n = f^{(n)}$; then

$$F_n(z) = \sum_{m=0}^{\infty} \xi_{n+m} \frac{\sqrt{(n+m)!}}{m!} z^m. \tag{3}$$

The common coefficient law has no atoms, so almost surely $\xi_k \neq 0$ for every k. On that event all Taylor coefficients in (2) are nonzero, and hence f is not a polynomial.

For later comparison with the literature, we record the growth of every sample. Applying Cauchy–Schwarz with weights $2^{k/2}$ gives

$$\begin{aligned} |f(z)| &\le \sum_{k\ge 0} \frac{|z|^k}{\sqrt{k!}} \\ &\le \left(\sum_{k\ge 0} \frac{2^k |z|^{2k}}{k!} \right)^{1/2} \left(\sum_{k \ge 0} 2^{-k} \right)^{1/2} = \sqrt{2}\exp(|z|^2). \end{aligned}$$

Thus (1) holds deterministically. In particular, the resulting function has order at most 2 and finite type in the usual terminology of entire-function theory.

We next establish the deterministic estimates behind the zero argument. For $r \ge 0$ set $a_{n,m}(r) = \sqrt{(n+m)!}\, r^m/m!$ and $S_n(r) = \sum_{m=0}^{\infty} a_{n,m}(r)$, and define the potential and loss

$$p_n(z) = \frac{1}{2}\log(n!) + |z|\sqrt{n}, \qquad L_n = 2 + \frac{1}{2}\log n. \tag{4}$$

The occurrence of $\log n$ below is always under hypotheses implying $n \ge 1$.

Lemma 2.1 (Saddle estimates)**.** *Let $n \ge 1$.*

(1) *If $r \ge 0$, $1 \le r\sqrt{n}$, and $r \le \sqrt{n}$, then for $m = \lfloor r\sqrt{n} \rfloor$,*

$$\log a_{n,m}(r) \ge \frac{1}{2}\log(n!) + r\sqrt{n} - 2 - \frac{1}{2}\log n.$$

(2) *For every $r \ge 0$,*

$$\log S_n(r) \le \frac{1}{2}\log(n!) + r\sqrt{n} + \frac{r+r^2}{2}.$$

Consequently, if $|\xi_k| \le 1$ and $|z| \le R$, then

$$\log|F_n(z)| - p_n(z) \le \frac{R+R^2}{2}, \tag{5}$$

with the inequality interpreted trivially if $F_n(z) = 0$.

Proof. Factor $a_{n,m}(r) = \sqrt{n!}\, b_{n,m}(r)$, where

$$b_{n,m}(r) = \frac{\sqrt{(n+1)(n+2)\cdots(n+m)}}{m!} r^m.$$

Writing $x = r\sqrt{n}$, we have $b_{n,m}(r) \ge x^m/m!$. For $m = \lfloor x \rfloor$, the assumptions give $1 \le m \le x \le n$. The elementary Stirling bound

$$\log(m!) \le m\log m - m + \frac{1}{2}\log m + 1$$

then yields, in one line,

$$\log\frac{x^m}{m!} \ge m\log\frac{x}{m} + m - \frac{1}{2}\log m - 1 \ge x - 2 - \frac{1}{2}\log n,$$

which proves (1).

For (2), use

$$b_{n,m}(r) = \sqrt{\frac{\binom{n+m}{n}}{m!}}\, r^m.$$

When $r > 0$, put $t = r/(r+\sqrt{n}) \in (0,1)$. Cauchy–Schwarz and the negative-binomial and exponential series give

$$\sum_{m\ge 0} b_{n,m}(r) \le \left(\sum_{m\ge 0} \binom{n+m}{n} t^m\right)^{1/2} \left(\sum_{m\ge 0} \frac{(r^2/t)^m}{m!}\right)^{1/2}$$
$$= (1-t)^{-(n+1)/2} \exp\left(\frac{r^2}{2t}\right).$$

Since $\log(1+u) \le u$ for $u \ge 0$ and $\sqrt{n} \ge 1$,

$$\frac{n+1}{2}\log\left(1+\frac{r}{\sqrt{n}}\right) + \frac{r(r+\sqrt{n})}{2} \le r\sqrt{n} + \frac{r+r^2}{2}.$$

The case $r = 0$ is immediate. Finally, (3) and the triangle inequality give $|F_n(z)| \le S_n(|z|)$, proving (5). ☐

The second ingredient is anti-concentration. Normalized area on $\overline{\mathbb{D}}$ satisfies, for $a \neq 0$, $w \in \mathbb{C}$, and $\varepsilon \ge 0$,

$$\mathbb{P}\{|a\xi + w| < \varepsilon\} \le \left(\frac{\varepsilon}{|a|}\right)^2. \tag{6}$$

Indeed, the preimage in $\overline{\mathbb{D}}$ is contained in a disk of radius $\varepsilon/|a|$, and the factor π cancels under normalization.

Lemma 2.2 (Pointwise logarithmic tail)**.** *Suppose* $1 \le |z|\sqrt{n}$ *and* $|z| \le \sqrt{n}$. *For* $s \ge 0$,

$$\mathbb{P}\left\{p_n(z) - \log|F_n(z)| > s + L_n\right\} \le e^{-2s}. \tag{7}$$

At a zero we assign the representative value $\log 0 = 0$. *This agrees with the Lean development and changes neither the pointwise probability, since the one-point law is nonatomic, nor any circle integral, since the zeros of a nonzero analytic function are isolated.*

Proof. Let $m = \lfloor |z|\sqrt{n} \rfloor$ and expose the single coefficient ξ_{n+m} in (3). Conditional on all other coefficients,

$$F_n(z) = a\xi_{n+m} + w, \qquad |a| = a_{n,m}(|z|).$$

Lemma 2.1(1) says $\log|a| \ge p_n(z) - L_n$. The event in (7) is therefore contained in $\{|a\xi_{n+m} + w| < |a|e^{-s}\}$. Apply (6) conditionally and integrate over the other coefficients. ☐

We need a circle version of this estimate. The following elementary form of the event-dependent logarithmic argument is convenient.

Lemma 2.3 (Circle lower tail)**.** *Let* $Y(\omega,\theta) \ge 0$ *be jointly measurable for* $0 \le \theta \le 2\pi$, *and suppose*

$$\mathbb{P}\{Y(\cdot,\theta) > s\} \le e^{-s/2} \quad (s \ge 0)$$

uniformly in θ. *If* $\overline{Y} = (2\pi)^{-1}\int_0^{2\pi} Y(\cdot,\theta)\,d\theta$, *then for* $t \ge 8$,

$$\mathbb{P}\{\overline{Y} \ge t\} \le 2e^{-t/8}.$$

Proof. The tail hypothesis implies integrability. If $E = \{\overline{Y} \ge t\}$ and $q = \mathbb{P}(E)$, Tonelli gives

$$tq \le \frac{1}{2\pi}\int_0^{2\pi} \mathbb{E}\big(Y(\cdot,\theta)\mathbf{1}_E\big)\,d\theta.$$

For any event of probability q, the layer-cake formula gives, uniformly in θ,

$$\mathbb{E}\big(Y(\cdot,\theta)\mathbf{1}_E\big) \le \int_0^\infty \min\{q, e^{-s/2}\}\,ds = 2q\log\frac{e}{q}.$$

If $q > 0$, then $t \le 2\log(e/q)$, so $q \le e^{1-t/2} \le 2e^{-t/8}$ for $t \ge 8$. The case $q = 0$ is trivial. ☐

3. Hole probabilities and extraction

For $c \in \mathbb{C}$ and $r > 0$, define

$$\gamma(c,r) = \frac{1}{2\pi}\int_0^{2\pi} |c + re^{i\theta}|\, d\theta - |c|.$$

Strict convexity of the Euclidean norm gives

$$\gamma(c,r) > 0. \tag{8}$$

Indeed, for every θ the triangle inequality gives $|c+re^{i\theta}| + |c - re^{i\theta}| \geq 2|c|$. The inequality is strict at some θ: this is immediate if $c = 0$, while for $c \neq 0$ one may choose $re^{i\theta}$ perpendicular to c. By continuity it is strict on an interval; integrating and shifting θ by π proves (8).

Fix $c \neq 0$ and $0 < R < |c|$, and put $r = R/2$. The circle $|z - c| = r$ lies in an annulus $\delta \leq |z| \leq M$ for some $\delta > 0$ and $M < \infty$. For all sufficiently large n, the hypotheses of Lemma 2.2 hold uniformly on this circle. Define

$$D_{n,z} = \max\{p_n(z) - \log|F_n(z)| - L_n, 0\}, \qquad \overline{D}_n = \frac{1}{2\pi}\int_0^{2\pi} D_{n,c+re^{i\theta}}\, d\theta.$$

The pointwise tail in Lemma 2.2 is stronger than the hypothesis of Lemma 2.3, so the latter applies to $\overline{D}_n$. The logarithms are locally integrable on each circle because F_n is analytic and not identically zero almost surely; alternatively, integrability follows directly from the uniform pointwise tail. Joint measurability follows from normal convergence of the random series and measurable finite partial sums.

Let $H_n(c,R)$ be the set of samples for which F_n has no zero in $B(c,R)$. On this set F_n is zero-free on $\overline{B(c,r)}$, and Jensen's formula [4, Chapter III] gives

$$\frac{1}{2\pi}\int_0^{2\pi} \log|F_n(c+re^{i\theta})|\, d\theta = \log|F_n(c)|. \tag{9}$$

By Lemma 2.1(2),

$$\log|F_n(c)| \leq p_n(c) + C(c), \qquad C(c) = \frac{|c| + |c|^2}{2}.$$

Since $D_{n,z} \geq p_n(z) - \log|F_n(z)| - L_n$, equations (8) and (9) imply on $H_n(c,R)$ that

$$\begin{aligned}\overline{D}_n &\geq \frac{1}{2\pi}\int_0^{2\pi} p_n(c+re^{i\theta})\, d\theta - p_n(c) - C(c) - L_n \\ &= \gamma(c,r)\sqrt{n} - C(c) - 2 - \frac{1}{2}\log n =: T_n.\end{aligned}$$

The measurable set $G_n = \{\overline{D}_n \geq T_n\}$ contains $H_n(c,R)$. Now $T_n \to \infty$ and, more precisely, $T_n \geq \frac{1}{2}\gamma(c,r)\sqrt{n}$ for all sufficiently large n. Writing $\mathbb{P}^*$ for outer probability, Lemma 2.3 therefore gives

$$\mathbb{P}^*\big(H_n(c,R)\big) \leq \mathbb{P}(G_n) \leq 2e^{-T_n/8} \leq 2\exp\left(-\frac{\gamma(c,r)}{16}\sqrt{n}\right) \tag{10}$$

for all sufficiently large n. The series of the right-hand side converges; for example, $\log n = o(\sqrt{n})$ gives an eventual comparison with n^{-2}. Hence

$$\sum_{n=0}^{\infty} \mathbb{P}^*\big(H_n(c,R)\big) < \infty. \tag{11}$$

This formulation avoids a separate measurability argument for the zero-free sets. The usual proof of the first Borel–Cantelli estimate [5, Section 2.3] applies verbatim to outer probability, since $\mathbb{P}^*(\limsup_n H_n) \leq \inf_N \sum_{n\geq N} \mathbb{P}^*(H_n) = 0$ whenever $\sum_n \mathbb{P}^*(H_n) < \infty$.

Choose a countable family $\mathcal{B}$ of disks $B(c,R)$ with $c \in \mathbb{Q} + i\mathbb{Q}$, $R \in \mathbb{Q}_{>0}$, and $R < |c|$. This is still an inner base for $\mathbb{C}$: every nonempty open set contains a nonzero point, then a nearby nonzero rational point and a sufficiently small rational disk around it. For each $B \in \mathcal{B}$, (11) and the first Borel–Cantelli lemma show that outside an outer-null set only finitely many F_n are zero-free in B. The countable union of these exceptional outer-null sets is outer-null. Intersecting its complement with the probability-one event that every ξ_k is nonzero gives a nonempty event on which all conclusions hold simultaneously.

Select a sample in this event and denote its entire function by f. It is nonzero and nonpolynomial. If $U \subset \mathbb{C}$ is nonempty and open, choose $B \in \mathcal{B}$ with $B \subset U$. All sufficiently high derivatives of f have a zero in B, hence in U. This proves Theorem 1.2; Lemma 1.1 supplies the equivalent subsequence statement.

4. Relation to a theorem of Boas and Reddy

Boas and Reddy state in [2, Theorem 1] that if an entire function has order at most 2 and finite type, then there is an arbitrarily large disk, somewhere in the plane, on which infinitely many of its successive derivatives are zero-free. The paragraph on the following page explicitly describes this as one fixed disk that is zero-free for infinitely many derivatives. Their expanded article is [3].

That assertion is incompatible with Theorem 1.2, not merely with a stronger uniform-density statement. Indeed, let D be any fixed nonempty open disk. Theorem 1.2 gives an $N(D)$ such that $f^{(n)}$ has a zero in D for every $n \geq N(D)$. Hence only finitely many derivatives can be zero-free on D. On the other hand, (1) places this same function within the growth class of the printed Boas–Reddy assertion. We conclude:

Theorem 4.1. *Theorem 1 of* [2] *is false as printed.*

The proof above is independent of the Boas–Reddy papers. The accompanying Lean theorem packages the cofinite conclusion and (1) for the same witness, so the correction does not depend on an informal identification of two separately constructed functions.

Formal verification. A machine-checked Lean 4 formalization of the main theorem and the supporting exact iterated-derivative formula, saddle estimates, small-ball and logarithmic-tail bounds, Jensen argument, summable hole probabilities, countable-base reduction, and deterministic extraction is available at

`github.com/erichou1/cofinite-derivative-zeros`.

The expanded public theorem is

```
CofiniteDerivatives.
exists_transcendental_entire_with_
explicit_derivative_zeros_
and_growth
```

The project is pinned to Lean 4.28.0 and Mathlib v4.28.0 [7]. The commands

`lake build CofiniteDerivatives` and `lake env lean Audit.lean`

check the build and print the theorem's axiom dependencies. The project-owned sources contain no `sorry`, `admit`, or custom axiom.

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References

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